Estimating Pedestrian Flows and Signal Delay from Sydney Traffic-Signal Event Histories

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Abstract

Traffic-signal histories record first pedestrian calls but omit subsequent arrivals. We apply a calibrated first-actuation estimator to Sydney histories for 18 September 2018. Screening retains 5,114 movement-hours at 174 sites, yielding an estimated 331,729 crossings. Estimated delay is 3,671.5 pedestrian-hours assuming full compliance and 2,746.3 hours under a transferred behaviour scenario: 39.8 and 29.8 seconds per crossing. Median site flow is 66.5 crossings per usable hour. The estimates depend on calibration transfer and screened coverage; they do not constitute a pedestrian census.

Keywords: pedestrian flow; pedestrian delay; traffic-signal histories; pedestrian actuation; censored estimation

1 Questions

The Sydney Coordinated Adaptive Traffic System (SCATS) records a pedestrian service request but generally adds no count for subsequent arrivals before Walk [Transport for NSW, 2026]. Earlier studies relate controller activity to observed pedestrian volumes [Singleton and Runa, 2021, Kothuri et al., 2024, Vahedi Saheli et al., 2026]; timing-based estimation instead uses signal events and an arrival model [Li and Wu, 2021]. We apply a separately calibrated first-actuation estimator [Levinson, 2026] to ask how estimated crossings and delay vary across Sydney sites, between full-compliance and behaviour-adjusted scenarios, and over the calibration-factor distribution.

2 Methods

2.1 Data

The archive contains 12,000 regional histories covering 750 dates from 1 September 2016 through 20 September 2018, with 229 intersections in 16 control regions. This application uses Tuesday, 18 September 2018. Timestamped demand, flashing-don't-walk and Walk events yield 9,626 candidate movement-hours at 187 sites.

A pedestrian movement is a controller crossing identifier. Unique spatial matches identify crosswalk orientation; counts combine both travel directions within a crosswalk. Individual travel direction is unobserved.

2.2 Method

For restrictive cycle i in movement-hour h, Y_i is latency from flashing-don't-walk onset to first actuation and C_i is observed no-actuation exposure. Let $\mathcal{A}_h$ and $\mathcal{C}_h$ denote actuated and censored cycles, $K_h = |\mathcal{A}_h|$, and T_h the clock-time exposure over which events are counted. The Poisson waiting-time estimator is

$$\begin{aligned} L_h &= \sum_{i \in \mathcal{A}_h} Y_i + \sum_{i \in \mathcal{C}_h} C_i, \quad \widehat{\lambda}_h = K_h / L_h, \\ \widehat{N}_h^e &= T_h \widehat{\lambda}_h, \qquad\qquad\qquad \widehat{N}_h = \widehat{\alpha}_N \widehat{N}_h^e. \end{aligned} \tag{1}$$

Here L_h is likelihood exposure, $\widehat{\lambda}_h$ is an event rate, $\widehat{N}_h^e$ is an event count, and $\widehat{N}_h$ estimates crossings.

Let R_i be first-actuation-to-Walk wait. Conditional on R_i and a constant rate, later events number $\widehat{\lambda}_h R_i$ in expectation and wait $R_i/2$ on average:

$$\begin{aligned} \widehat{D}_i &= R_i + \widehat{\lambda}_h R_i^2/2, \quad \widehat{D}_h = \sum_{i \in \mathcal{A}_h} \widehat{D}_i, \\ \widehat{D}_{h,100} &= \widehat{\alpha}_{D,100} \widehat{D}_h, \qquad \widehat{D}_{h,b} = \widehat{\alpha}_{D,b} \widehat{D}_h. \end{aligned} \tag{2}$$

Both terms estimate event-equivalent delay. Separate empirical factors convert counts to crossings and complete delay totals to pedestrian-seconds for each target. Applying the count factor only to later arrivals would change the calibrated estimator (SI-2).

The transferred factors are $\widehat{\alpha}_N = 2.203$, $\widehat{\alpha}_{D,100} = 2.212$ and $\widehat{\alpha}_{D,b} = 1.655$ [Levinson, 2026]. Empirical case-level ranges describe transfer sensitivity. Crossings are held fixed between delay scenarios; $\widehat{\kappa} = \widehat{\alpha}_{D,b}/\widehat{\alpha}_{D,100}$. Local compliance and delays before button pressing are unobserved, and calibration transfer is unverified at these sites (SI-7).

Movement-hours require at least five calls, less than 50% zero-latency calls, less than 75% no-actuation cycles, and no excluded controller pattern (SI-3). Missing estimates are not zeros. Adjacent-hour pooling could improve sparse-call support under a common-rate assumption; it is not applied here.

We sum retained movements to sites. Flow divides crossings by hours containing at least one retained movement; average delay divides total delay by crossings. Site 516's weekly comparison holds movement and clock-hour coverage constant across all seven days. Monthly estimates pool matching movements and clock hours across dates, retaining common coverage across September 2017–August 2018 and excluding two incomplete dates (SI-4).

3 Findings

Statistical screens exclude 4,290 candidate movement-hours and controller-pattern screens remove another 222, leaving 5,114 at 174 sites. Ten terminal actuations without subsequent Walk contribute to counts but not delay. Table 1 reports 331,729 estimated crossings and the calibration-transfer range.

Figure 1 maps site flow and the four crosswalks at site 516 (Willoughby Road/Albany Street, Crows Nest). The inventory's rounded coordinates place multiple sites at identical map positions; display offsets distinguish them. Site 516 contributes 6,095.4 crossings and 39.7 pedestrian-hours of delay over 17 usable hours, or 358.6 crossings/hour.

Table 1: Count coefficient from Levinson [2026] and network estimate. The factor and crossing ranges propagate the reported calibration-factor distribution.

Calibration source	Mean $\widehat{\alpha}_N$	Factor range	Estimated crossings	Crossing range
Separate calibration study	2.203	1.417–3.138	331,729	213,384–472,525

Figure 2 shows site 516's temporal estimates. The main-day peak is 659.1 crossings at 13:00; overnight gaps reflect sparse or absent qualifying records under the screens, independent of calibration clock-hour coverage (SI-3). Comparable daily estimates range from 4,162 on Sunday to 6,694 on Friday; monthly typical-day estimates range from 5,132 in June to 6,102 in September.

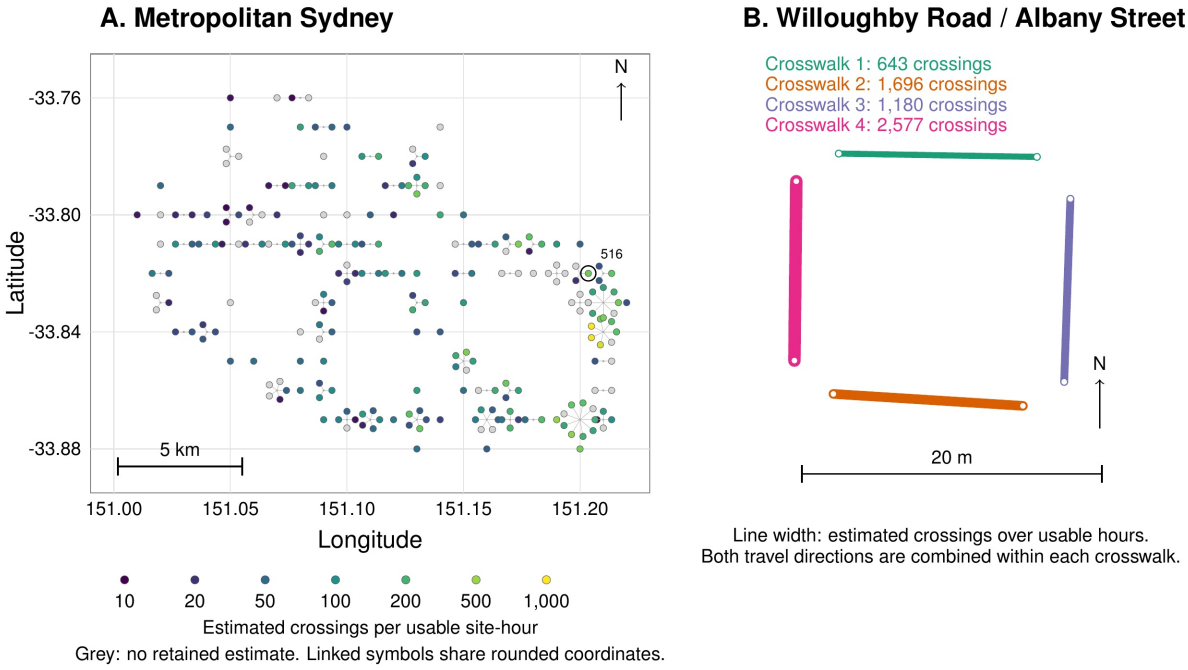


Figure 1: Estimated pedestrian activity on 18 September 2018. Panel A shows the 229-site inventory, with colour indicating estimated crossings per usable site-hour at 174 retained sites; grey symbols indicate sites without retained estimates. The inventory coordinates are rounded to 0.01°. Sites sharing a coordinate are separated by small display offsets and linked to that coordinate; offsets do not indicate precise site locations. The outlined symbol identifies site 516. Panel B shows its four crosswalks; line width indicates estimated crossings over usable hours, combining both travel directions.

Table 2 gives 3,671.5 pedestrian-hours assuming full compliance and 2,746.3 under the behaviour-adjusted scenario, or 39.8 and 29.8 seconds per crossing. The reduction is 925.2 hours (25.2%); it is a scenario comparison, not an observed compliance effect.

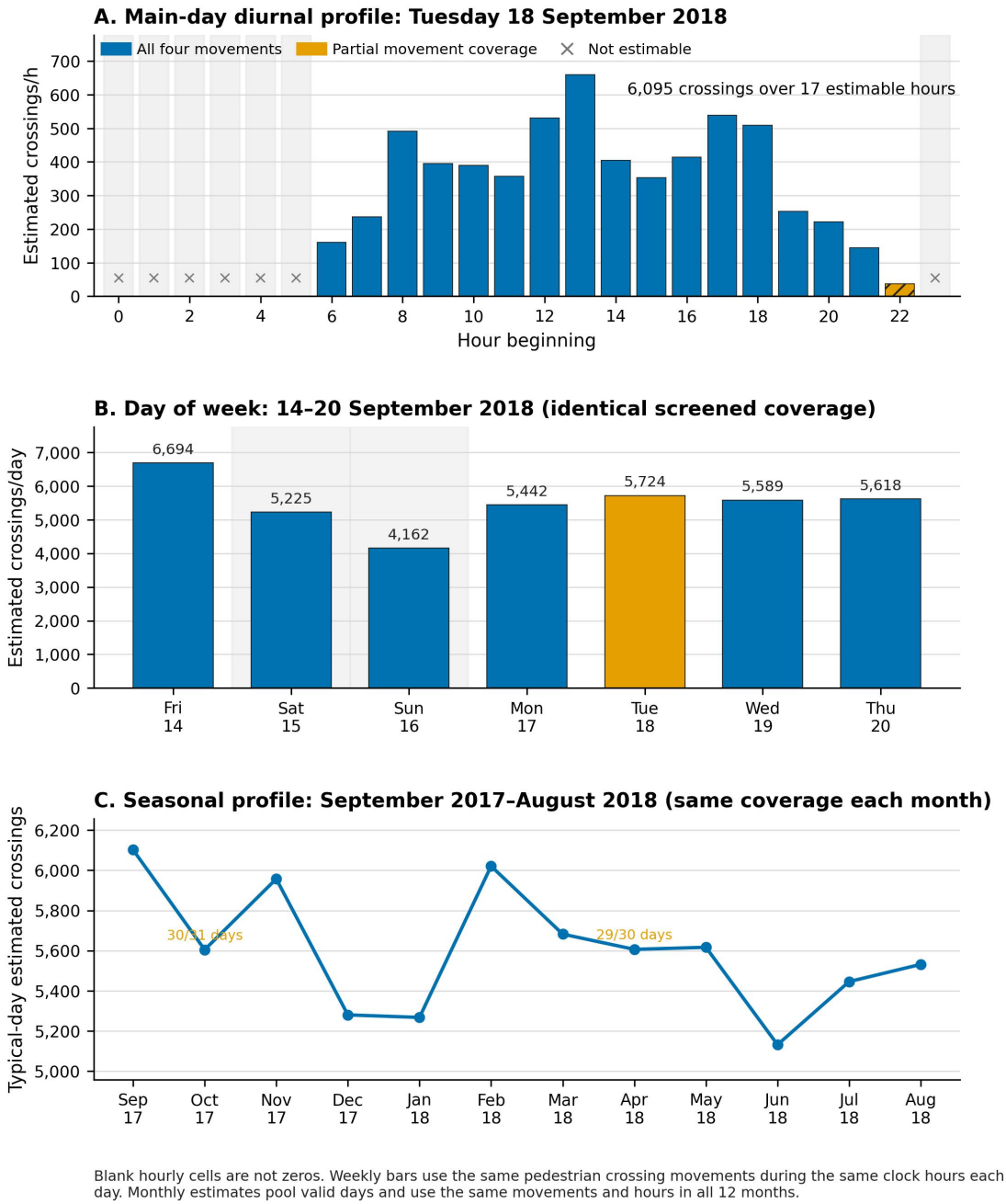


Figure 2: Estimated pedestrian crossings at site 516. Panel A shows the 24-hour profile for 18 September 2018. Crosses mark hours that are not estimable under the screening rules; the hatched 22:00 bar uses three of the four pedestrian movements. Panel B compares seven consecutive days using identical screened movement and clock-hour coverage; the Tuesday analysis date is highlighted. Panel C shows monthly typical-day estimates from September 2017 through August 2018 using the same movements and clock hours in every month. Orange labels identify months in which one incomplete date is excluded. Monthly values are typical-day estimates, not monthly totals.

Table 2: Estimated network delay using the reported calibration coefficients, assuming 100% compliance and under the behaviour-adjusted scenario. Estimated crossings are held fixed as an analytical control.

Delay target	Mean $\widehat{\alpha}_D$	Factor range	Total (h)	Total range (h)	s/crossing	$\widehat{\kappa}$
Assuming 100% compliance	2.212	1.217–3.671	3,671.5	2,020.3–6,092.2	39.8	1.000
Behaviour-adjusted	1.655	0.886–2.738	2,746.3	1,470.9–4,544.9	29.8	0.748

Table 3 reports median site flow of 66.5 crossings/hour (interquartile range 32.9–150.7) and median average delay of 48.9 seconds/crossing (39.6–59.9). Figure 3 displays the four distributions and calibration-transfer envelopes. Sites contribute 1–22 usable hours (median 15), totalling 2,307. Crossing-weighting gives the lower network average delay of 39.8 seconds. The behaviour-adjusted site median is 36.6 seconds. Crosswalk orientation is identified for 113 movements at 61 sites; all retained movements contribute to network totals.

Table 3: Distributions across 174 sites using the reported mean calibration coefficients and assuming 100% compliance

Site metric	Median	Interquartile range
Estimated crossings per usable site-hour	66.5	32.9–150.7
Estimated crossings over usable hours	927	394–2,414
Average delay (s/crossing)	48.9	39.6–59.9
Total delay (pedestrian-hours)	12.9	5.2–26.9

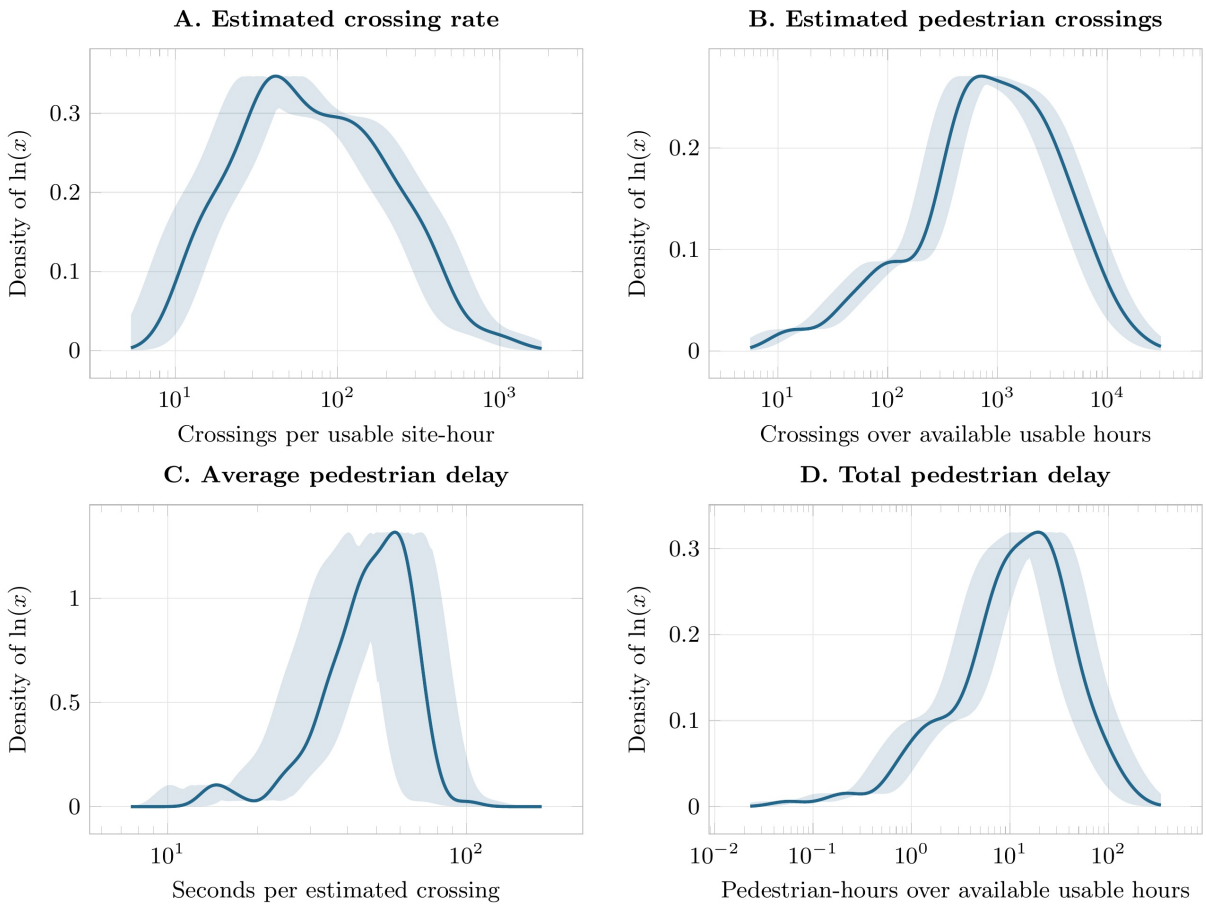


Figure 3: Site probability densities for estimated crossing flow, crossing count, average delay, and total delay, with delay estimated assuming 100% compliance. The line uses the reported mean factor estimates; the ribbon propagates the reported calibration-factor distribution. Supplemental Information describes the transfer calculation.

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Data and Code Availability. Raw SCATS controller histories cannot be redistributed under the archive terms. On acceptance, reproducible scripts and derived, nonidentifying movement-hour, site, calibration-factor, and figure-source outputs will be deposited in a public GitHub repository, and the repository URL will be added to the final article.

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